

PSYC 300A · MIDTERM 2 CHEAT SHEET

CORRELATION & REGRESSION · $\neq$ NOT ON THE PROF'S PROVIDED SHEET – KNOW THESE

① CORE FORMULAS (BUILD-UP)

▮ **Mean:** $\bar{x} = \Sigma x / N$

▮ **Sum of Squares:** $SS = \Sigma (x - \bar{x})^2$

▮ **Variance:** $SD^2 = SS / N$

▮ **Std Dev:** $SD = \sqrt{(SS / N)}$

$SP = \Sigma (x - \bar{x})(y - \bar{y})$

$COV = SP / N$

$r = COV / (SD_x \cdot SD_y)$

Do SS & SD for X and Y separately.

② CALCULATION ORDER (THE TABLE)

One table, columns:

$X \cdot Y \cdot (x - \bar{x}) \cdot (y - \bar{y}) \cdot (x - \bar{x})^2 \cdot (y - \bar{y})^2 \cdot (x - \bar{x})(y - \bar{y})$

Sum the last 3 cols → **SS_x , SS_y , SP** . Then SD_x , SD_y , COV , r .

Round 4 dp during, 2 dp final.

③ DESCRIBE A CORRELATION

- **Direction** + / – / none
- **Shape** linear / curvilinear
- **Strength** weak/mod/strong (subjective)
- **Magnitude** = r value (–1 to +1; objective)

In words = 3 terms: **strength + direction + shape** (e.g. "strong, negative, linear").

④ INTERPRET R

Human behaviour r 	=
< .10	none
.10–.30	weak
>.30–.50	moderate
> .50	strong
Reliability r 	=
> .85	very desirable
.70–.85	acceptable
< .70	poor

⑤ WHICH CORRELATION? (BY SCALE)

	Y Nom	Y Ord	Y I/R
X Nom	Phi ϕ	Spear	Pt-Bis
X Ord	Spear	Spear	Spear
X I/R	Pt-Bis	Spear	Pearson

- Both interval/ratio → **Pearson r**
- Any ordinal → **Spearman r_s**
- 1 dichot-nominal + I/R → **Point-Biserial r_{pb}**
- Both dichot-nominal → **Phi ϕ**

Also use Spearman if I/R data is **curvilinear** or **heteroscedastic** (ranks the scores, pulls outliers in).

⑥ FACTORS AFFECTING R

Does NOT affect r : sample size (N cancels).

1. Real strength of relationship
2. Sampling error ($\uparrow$ or $\downarrow r$)
3. 3rd variable (corr $\neq$ cause)
4. Heterogeneous sample (subgroups)
5. Restricted/truncated range
6. Non-linearity → r **underestimates**
7. Heteroscedasticity → $SD \uparrow$ so $r \downarrow$

REGRESSION — predict Y' from X

SAME ENGINE AS R, TAKEN ONE STEP FURTHER · π = NOT ON PROVIDED SHEET

⑦ REGRESSION FORMULAS (PROVIDED)

Slope: $b_y = r \cdot (SD_y / SD_x)$

Intercept: $a_y = \bar{y} - b_y(\bar{x})$

Line: $Y' = a_y + b_y(x)$

 $b_x = r \cdot (SD_x / SD_y)$

$a_x = \bar{x} - b_x(\bar{y})$

$X' = a_x + b_x(y)$

⑧ REGRESSION PIPELINE

1. Scatterplot + mean & SD each var
2. Pearson r
3. $b_y = r \cdot SD_y / SD_x$
4. $a_y = \bar{y} - b_y(\bar{x})$
5. $Y' = a_y + b_y(x)$ → plug in X
6. Plot via (0, a_y) and ($\bar{x}$, $\bar{y}$)

X' : same steps, swap X & Y.

⑨ PREDICTING — KEY RULES

- Predictor = X (predict from). Criterion = Y (predicted).
- **NO extrapolation:** if the value is outside the observed range → can't predict → write **NA**.
- Plug into Y' (or X'); show formula → numbers → answer.

⑩ TWO LINES: Y' & X'

- Cross at the means ($\bar{x}$, $\bar{y}$) (if not → error).
- Rotate around ($\bar{x}$, $\bar{y}$) as r changes.
- Small angle = high r. $r = \pm 1$ → lines **overlap**. $r = 0$ → **perpendicular** (90°).

⑪ R, B, A RELATIONSHIP

- $r = 0$ → $b = 0$; Y' flat at $a_y = \bar{y}$; X' vertical at $a_x = \bar{x}$.
- $r \uparrow$ → $|b| \uparrow$ (steeper).
- $b +$ → a below mean (toward origin); $b -$ → a above mean.

⑫ STD ERROR OF ESTIMATE

$SD_{(Y-Y')} = SD_y \cdot \sqrt{(1 - r^2)}$

$= \sqrt{(\sum (Y-Y')^2 / N)}$

- $r \neq 0$ → smaller than SD_y
- $r = 0$ → equals SD_y
- $r = \pm 1$ → = 0 (no error)

⑬ VARIABILITY (R^2)

Total $(Y-\bar{y}) =$ Explained $(Y'-\bar{y}) +$ Unexpl. $(Y-Y')$

$SS_T = SS_R + SS_E$

- $r^2 = SS_R / SS_T =$ proportion **explained**
- $1 - r^2 = SS_E / SS_T =$ proportion **unexplained**

e.g. $r=.95 \rightarrow r^2=.90 \rightarrow 90\%$ explained.

⑭ KEY CONCEPTS (1-LINERS)

- **Least squares:** best line minimizes $\sum (Y-Y')^2$.
- **Heteroscedasticity** (links both topics): Y-spread changes across X (fan). $\downarrow r$, flattens slope → use Spearman.
- **Curvilinear:** Pearson underestimates → Spearman.
- **Restricted range:** can't generalize beyond sampled range.
- **Corr $\neq$ cause:** need an experiment.

⑮ EXAM REMINDERS

- Every calc: **formula** → **numbers** → **answer** (2 dp).
- Round 4 dp during, 2 dp final.
- Write **NA** if a value shouldn't be computed.
- Scientific (non-graphing) calculator.